Kento Nishi

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Education

Massachusetts Institute of Technology — PhD program in EECS Sep. 2026 – Present

- CSAIL (Computer Science & Artificial Intelligence Laboratory). Research advisor: Phillip Isola.

Harvard University — College & SEAS Aug. 2022 – May 2026

- Concurrent four-year Bachelor's & Master's; Honors cum laude AB/SM in CS. GPA: 3.976(U), 4.000(G).

Skills

Programming: Python, TypeScript/JavaScript, C++, Java, Svelte, OpenGL, Bash, Git, Docker, LaTeX.
Research: PyTorch, CNNs, transformers, diffusion, mechanistic interpretability, representation learning, explainability, multi-modal learning, data augmentation, LLMs, audio processing, full-duplex models.
Misc.: bilingual English/Japanese (professional-level), teaching, distance running, music composition.

Affiliations

CSAIL @ **MIT EECS**, w/ P. Isola. Sep. 2026 – Present
Physics of AI @ **Harvard-NTT**, w/ H. Tanaka & E. S. Lubana. Mar. 2023 – May 2026
Visual Computing Group (VCG) @ **Harvard**, w/ H. Pfister. Sep. 2022 – May 2026
Four Eyes Lab @ **UC Santa Barbara**, w/ T. Höllerer. Jun. 2020 – Aug. 2022

Grants & Fellowships

Ezoe Memorial Recruit Foundation Scholarship Apr. 2026 – Present

- Up to $95,000/year, covering tuition and living expenses; renewed through first PhD year.
- Oldest & most selective scholarship program in Japan (approx. 6 recipients/year).

MIT EECS Distinguished Graduate Fellowship Sep. 2026 – Present

- Jae S. Lim Graduate Fellowship Fund.

Experience

Comcast — Applied AI Research, Speech AI Team

PhD Research Intern (full-time, remote) Jun. – Aug. 2026

- Mitigated speech timing issues in full-duplex models (Moshi and PersonaPlex). Patent underway.
- "Causal Analysis and Mitigation of Spurious Onsets in Full-Duplex Speech LLMs" for ICASSP 2027.

Research Contractor (part-time remote internship continuation) Sep. – Dec. 2025

- Continued post-internship research on multimodal speech and diffusion-based speech generation.

Master's Research Intern (Washington DC, in-person) Jun. – Aug. 2025

- Developed production-scale speech models serving over 50 million daily users on Xfinity & Sky TV.

Harvard SEAS

CS79 Teaching Fellow, Design of Useful and Usable Interactive Systems Fall 2025

- Sole undergraduate member of course staff; studio planning lead; mentored approx. 12 students.

Distinctions

Tier-1 Conference Reviewer: ICLR 2026 (invitation-only). 2025
John Harvard Scholar: top 5% GPA in the Harvard College class of 2026. 2023, 2025
Sub-3 Marathoner: 2h56m19s PR (Santa Rosa, Aug. 2026). 1h21m29s half (Cambridge, Nov. 2025).
Regeneron STS Top 300 Scholar: the oldest, most prestigious high school STEM award. 2022

Research Themes

Science of deep learning

- Developed a mechanistic account of physical misgeneralization in sequence models for NeurIPS 2026.
- Worked on pre-training & in-context mechanisms w/ interpretable synthetic graph tasks (2023–2025).
- **First-author ICML 2025**; co-author **ICLR 2025** & **ICML 2024**; **first-author ICLR 2026** and co-author **NeurIPS 2023 & 2024** workshops.
- Supported by Harvard PRISE 2024 ($3K + residency) & CBS-NTT Fellowship 2023 ($5K).
- Advised by H. Tanaka & E. S. Lubana (PAI), H. Pfister (VCG), and K. Kumar & R. Tang (Comcast).

Multimodal modeling (audio & speech; computer vision)

- Analyzed spurious onsets in full-duplex speech systems at Comcast. Submitting to ICASSP 2027.
- Worked on multimodal and diffusion-based speech generation models (2025).
- Enhanced robust vision under partial/noisy supervision (2020–2024); **first-author CVPR 2024**.
- **First-author CVPR 2021 & AAAI 2021** at age 16.
- Advised by K. Kumar & R. Tang (Comcast), H. Pfister (VCG), and T. Höllerer (Four Eyes Lab).

AI governance, safety, and scientific applications

- **MATS 7.0** (winter 2025) research fellow; $12K grant + $10K compute.
- Mixed-methods study of ethics review across AI paper resubmissions for NeurIPS 2026 workshop.
- "Evolutionary Curriculum Learning Improves Biological Sequence Modeling," **ICML 2026** workshop.

Technical Projects

LiveTL Apps

- Founding dev of LiveTL, HyperChat, and YtcFilter, addons that improve YouTube & Twitch.
- 130K+ total users; 1,000+ repo stars; 30+ code contributors. Free, open-source, and cross-platform.

Torch Pitch Shift — PyPI

- The first Python library for pitch-shifting on GPU at the time; later added to PyTorch upstream.
- 1.7M+ downloads/month; 135+ GitHub stars; used by torch-audiomentations (1.2K+ stars).

Torch Time Stretch — PyPI

- GPU-accelerated PyTorch library for fast audio time-stretching and efficient transformation search.
- 40+ GitHub stars; used by ByteDance's BuboGPT research code (500+ stars).

iframe Translator — npm

- A reverse-engineered hack of Google's free in-browser translation feature, exploiting iframe loopholes.

slsh: SSH without keyboard lag

- Built a drop-in SSH wrapper that hides interactive latency using temporary local terminal predictions.
- Written in Rust; supports Linux, macOS, and Windows on x86-64 and ARM64.

holoEN Christmas Advent Calendar

- Full-stack dev of the event platform (holoen-advent.com); officially commissioned by Cover Corp.
- 250K+ viewers; a beloved yearly tradition in the hololive English community (2022, 2023, 2024).

Sound & Audio Technology

Music composition/production

- Making original music since 2017; 20+ tracks w/ Reaper, VSTs, MIDI controllers, etc.
- Highlights: "HyperChat Trailer Theme" (2022, 16K views on YouTube), "Bliss" (2026, YouTube).

Plugins & tools

- Pogify: synchronized music playback extension. 10+ contributors; 50+ stars; 15K+ social media likes.
- ClipSynth: WebAssembly, Web Audio, WebUSB, Web MIDI tool to convert arbitrary clips into synths.

All Publications

Mechanisms of Misgeneralization in Physical Sequence Modeling
Kento Nishi, Raphael Tang, Karun Kumar, Core Francisco Park, Hidenori Tanaka
arXiv Preprint 2026, as **first author**.
- We show generated trajectories can look plausible while shifting aggregate distance or energy.
- We use a data deviation kernel to predict and reduce this form of physical misgeneralization.

Evolutionary Curriculum Learning Improves Biological Sequence Modeling
Richard Zhu, Kento Nishi
ICML 2026 SPIGM Workshop, as co-author.
- We gradually expand training from close evolutionary neighbors to distant homologs.
- This improves protein variant-effect prediction and RNA sequence generation.

When does Observational Data Teach Latent Dynamics? Understanding Control Misalignment with Synthetic Tasks
Kento Nishi, Raphael Tang, Karun Kumar, Core Francisco Park, Hidenori Tanaka
ICLR 2026 Sci4DL Workshop, as **first author**.
- We show generated samples can fit data while violating latent controls like speed or energy.

Representation Shattering in Transformers: A Synthetic Study with Knowledge Editing
Kento Nishi, Rahul Ramesh, Maya Okawa, Mikail Khona, Hidenori Tanaka, Ekdeep Singh Lubana
ICML 2025, as **first author**.
- We show editing can distort untargeted entity representations and factual-recall geometries.

In-Context Learning of Representations
Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Yongyi Yang, Maya Okawa, Kento Nishi, Martin Wattenberg, Hidenori Tanaka
ICLR 2025, as co-author.
- We show concept representations can mediate learning from graph-structured examples.

Structured In-Context Task Representations
Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Kento Nishi, Maya Okawa, Hidenori Tanaka
NeurIPS 2024 NeurReps Workshop, as co-author.
- Precursor to ICLR 2025 "In-Context Learning of Representations."

Towards an Understanding of Stepwise Inference in Transformers: A Synthetic Graph Navigation Model
Mikail Khona, Maya Okawa, Jan Hula, Rahul Ramesh, Kento Nishi, Robert Dick, Ekdeep Singh Lubana, Hidenori Tanaka
ICML 2024, as co-author.
- We test stepwise inference, simplicity bias, and in-context primacy with graph navigation.

Stepwise Inference in Transformers: Exploring a Synthetic Graph Navigation Task
Mikail Khona, Maya Okawa, Rahul Ramesh, Kento Nishi, Robert P. Dick, Ekdeep Singh Lubana, Hidenori Tanaka
NeurIPS 2023 R0-FoMo Workshop, as co-author.
- Precursor to ICML 2024 "Towards an Understanding of Stepwise Inference in Transformers."

Joint-Task Regularization for Partially Labeled Multi-Task Learning
Kento Nishi, Junsik Kim, Wanhua Li, Hanspeter Pfister
CVPR 2024, as **first author**.
- We regularize shared-space outputs to improve partially labeled multi-task learning at linear cost.

Augmentation Strategies for Learning with Noisy Labels
Kento Nishi, Yi Ding, Alex Rich, Tobias Höllerer
CVPR 2021, as **first author**.
- We separate loss-modeling and learning augmentations to improve noisy-label robustness.

Improving Label Noise Robustness with Data Augmentation and Semi-Supervised Learning
Kento Nishi, Yi Ding, Alex Rich, Tobias Höllerer
AAAI 2021 Student Abstract Track, as **first author**.
- Precursor to CVPR 2021 "Augmentation Strategies for Learning with Noisy Labels."